The University of the South Pacific

School of Information Technology, Engineering, Mathematics and Physics

CS352 – Cybersecurity Principles

Semester 1, 2025

Mid-Semester TEST

Student ID: _____

Full Name: SOLUTION

Learning Outcomes

1. Explain the terminology and key principles associated with cybersecurity.

Possible Marks: 30
Weight: 10%
Time Allowed: 45 minutes
Pages: 6 (including this page)

Instructions

1. This test has two sections
   a. Section A – 8 marks
   b. Section B – 22 marks
2. All sections and questions are compulsory.
3. For all questions, use the space provided to write out your answers.

~ALL THE BEST~

SECTION A (8 Marks)

1. Define the following four terms and explain how they relate to cybersecurity: **(4 Marks)**

a) Vulnerability
A vulnerability is a weakness that can be exploited by cybercriminals to gain unauthorized access to a computer system.

b) Threat
A threat can exploit a vulnerability and can damage or destroy an asset.

c) Risk
The probability of loss due to a security incident.

"The effect of uncertainty on objectives" or the product of the severity of an event and the likelihood of it occurring.

d) Privacy
An individual's **right to control** how and to what extent information about him or her may be shared and acted upon by others.

2. Discuss why is it important to avoid using the same password for multiple accounts? **(2 Marks)**

Using the same password leaves you and your information vulnerable to financial and identity theft. If your password in one website is compromised hackers can use the same password to gain access to your other online accounts.

Therefore, it is important to use a unique one for each of your accounts.

3. Explain the concept of a security control within the realm of cybersecurity and information technology. **(2 Marks)**

The protection of information and information systems from **unauthorized access, use, disclosure, disruption, modification, or destruction** in order to provide confidentiality, integrity, and availability.

SECTION B (22 Marks)

4. Risk matrices are commonly used to display an organisation's risk profile. Despite this, they have been heavily critiqued in recent times.

a) Explain what a **risk matrix** is. **(2 Marks)**

A risk matrix is a likelihood consequences chart upon which risks can be placed to aid in prioritization based on the severity of a risk to an organisation.

b) Define a **risk profile**. **(1 Mark)**

The risk profile is a description of the risks of an enterprise, resulting from the risk assessment.

c) State ONE shortcoming or weakness of a risk matrix when used to showcase cybersecurity risks. **(1 Mark)**

- Some threats can be non-linear so displaying them here can under-or overestimate their severity
- Doesn't tell you HOW to respond to threats
- If on a diagonal doesn't tell you which threats to respond to first
- Consequence is a largely subjective measure
- Consequences and likelihood of a risk may not be independent

5. For an organisation such as USP, **state** whether you believe a HIDS or a NIDS is more important and why. **(3 Marks)**

HIDS = Host Intruder Detection System

NIDS = Network Intruder Detection System.

HIDS is important as staff, students and the public can directly or indirectly bring malware onto hosts at USP with little difficult (USB drives, email attachments)

NIDS is important because of how many campuses USP has giving them a broad attack surface. USP also has numerous valuable databases that are potential hacker targets. There is also a significant risk of insider threats here because of the level of education of staff and students and their access to the broader internet.

(either answer was fine – it just had to be justified appropriately. Obviously both are necessary for USP)

6. State an issue with just using log file analysis instead of an IDS or IPS to prevent cyberattacks. Then explain why even a good IDS and firewall cannot prevent all malicious traffic. **(3 Marks)**

Logfile analysis takes effort and time and familiarity with all systems beforehand. Logfiles are also riddled with false positives making manual inspection often worthless. Also logfiles often need to be collated in order for events to become apparent in many places.

Firewalls only block traffic that is EXPLICITLY malicious. Furthermore any encrypted payload will be able to make it through most IDSs and firewalls and as the content cannot be determined without first breaking the encryption.

7. **Prevention** is often discussed as the major component of a cyberdefence system. What are the other two major components that also need considering? Give an example for each component. **(4 Marks)**

Detective controls - monitor to receive warning of attempts to exploit vulnerabilities

e.g. intrusion detection system

Corrective controls - aim to correct errors or irregularities that have been detected

e.g. restoring to backup image

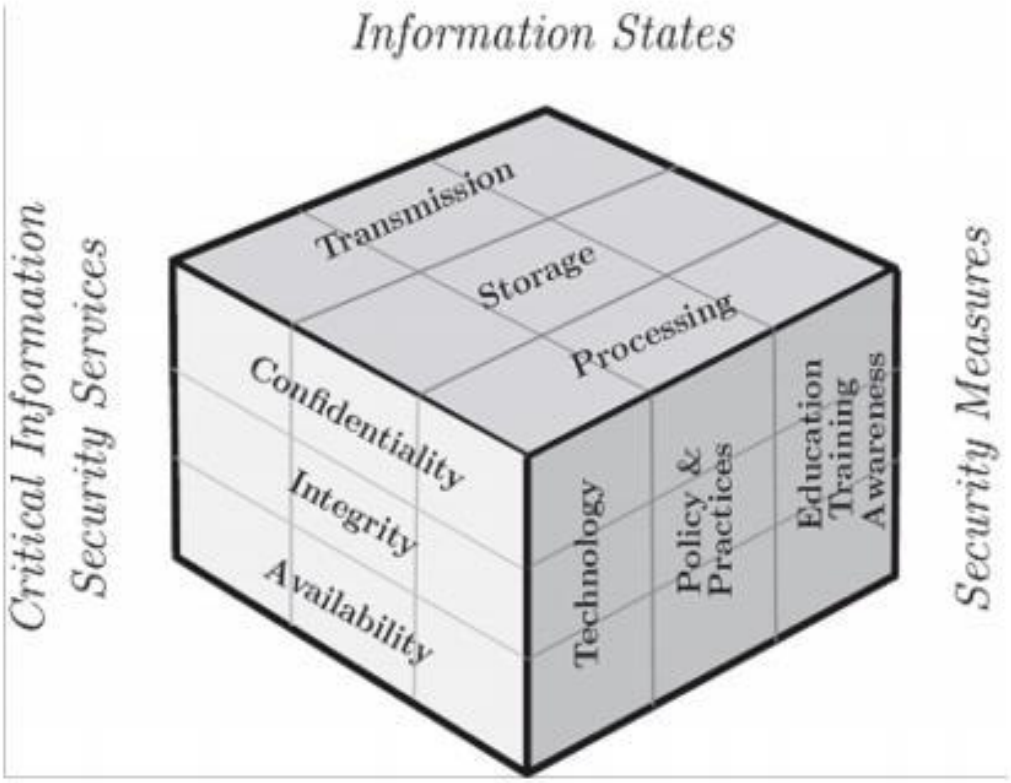
8. List the 3 major steps of a risk assessment? **(3 Marks)**

Risk identification

Risk Analysis

Risk Evaluation

9. Draw and discuss the **Information Security Model**. Ensure your answer includes the different components of the security model. **(5 Marks)**



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